NeuroPipeline DICOM Converter: DICOM to BIDS in one workflow

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Abstract

Keywords: [TODO: Keyword 1], [TODO: Keyword 2], [TODO: Keyword 3], [TODO: Keyword 4]

Table 1: Specifications Table.

Subject area	Neuroscience
Specific subject area	Multimodal human MRI (structural, functional, diffusion, and field-map imaging) with four study cohorts labelled in metadata as Control ($n = 56$), Glaucoma ($n = 19$), DataON ($n = 7$), and DataTON ($n = 2$) (84 participants total).
Type of data	Raw MRI images (NIfTI); BIDS metadata (JSON, TSV); task event files (where available); physiological recordings (where conversion gates were met); optional derivatives (defaced anatomicals; partial preprocessed DWI).
How the data were acquired	Siemens Prisma, 3 T MRI. Longitudinal design with up to two sessions (`ses-01`, `ses-02`). Functional paradigms present in the release include `task-rest`, `task-movie`, `task-fmri` (visual grating/checkerboard), and `task-control`. Conversion to BIDS with dcm2niix / `neuro_pipeline`.
Data format	BIDS (BIDSVersion 1.9.0); NIfTI (`.nii.gz`); JSON sidecars; TSV; physiology `.tsv.gz` + JSON (where present).
Data source location	[TODO: Institution / city / country — not stated in the current public release metadata (administrative institution fields were scrubbed from BIDS sidecars).]
Data accessibility	Repository: FITBIR (accession / URL [TODO: insert after deposit]). Access: restricted under FITBIR data-access agreement (see dataset `LICENSE`; not CC0).
Related research article	[TODO: Insert related research article citation, or state N/A if none.]

1. Background and Motivation

1.1. Background

1.2. Motivation

2. Data Description

This data release comprises multimodal magnetic resonance imaging (MRI) from a longitudinal visual-system study, organised according to the Brain Imaging Data Structure (BIDS) version 1.9.0 (`DatasetType: raw`). The packaged tree contains **84** participants (`sub-001`–`sub-084`) and **135** imaging sessions (`ses-01`: 83; `ses-02`: 52). Fifty-one participants have both sessions, 32 have `ses-01` only, and one participant (`sub-043`) has `ses-02` only; unavailable visits are omitted rather than represented by empty folders. Cohort labels in `participants.tsv` are Control ($n = 56$), Glaucoma ($n = 19$), DataON ($n = 7$; optic neuritis), and DataTON ($n = 2$; traumatic optic neuropathy); sex is recorded as Female ($n = 46$) and Male ($n = 38$). Demographic counts by cohort and session are summarised in Table 2. Released imaging modalities include anatomical MRI, blood-oxygen-level-dependent (BOLD) functional MRI, spin-echo field maps, and diffusion-weighted MRI, together with task event tables, physiological recordings, and selected derivatives. The overall organisation of the released dataset is illustrated in Fig. 1.

Table 2: Participant demographic characteristics by cohort and imaging session (age omitted). Each cohort has ses-01 and ses-02 columns. Values are n or n (%) as indicated.

Characteristic	Control		Optic neuritis (ON)		Traumatic optic neuropathy (TON)		Glaucoma		Total	
	ses-01	ses-02	ses-01	ses-02	ses-01	ses-02	ses-01	ses-02	ses-01	ses-02
Participants, n	55	41	7	1	2	1	19	9	83	52
Female, n (%)	35 (63.6%)	25 (61.0%)	6 (85.7%)	1 (100.0%)	0 (0.0%)	0 (0.0%)	5 (26.3%)	3 (33.3%)	46 (55.4%)	29 (55.8%)
Male, n (%)	20 (36.4%)	16 (39.0%)	1 (14.3%)	0 (0.0%)	2 (100.0%)	1 (100.0%)	14 (73.7%)	6 (66.7%)	37 (44.6%)	23 (44.2%)

At the dataset root, `dataset_description.json` records the dataset name, BIDS version, raw dataset type, licence statement, and conversion provenance (`GeneratedBy: neuro_pipeline` version 2.1.0). Participant-level metadata are limited to `participant_id`, `cohort`, and `sex` in `participants.tsv`, with column definitions in `participants.json`. Additional root files include `README.md`, `CHANGES`, and task metadata sidecars `task-movie.json`, `task-movie_events.json`, and `task-fmri_events.json`. A `code/` directory provides stimulus-related resources used to interpret functional events

```
release_dataset
├ dataset_description.json
├ participants.tsv / participants.json
├ README.md / CHANGES
├ task-movie.json / task-*_events.json
├ sub-XXX
│ └ ses-XX
│   ├ anat/
│   │ ├ sub-001_ses-01_run-01_T1w.nii.gz / .json
│   │ ├ sub-001_ses-01_run-03_T1w.nii.gz / .json (WMn)
│   │ ├ sub-001_ses-01_run-01_FLAIR.nii.gz / .json
│   │ └ sub-001_ses-01_run-01_TB1TFL.nii.gz / .json
│   ├ func/
│   │ ├ sub-001_ses-01_task-rest_run-01_bold.nii.gz / .json
│   │ ├ sub-001_ses-01_task-movie_run-01_bold.nii.gz / .json
│   │ ├ sub-001_ses-01_task-fmri_run-01_bold.nii.gz / .json
│   │ ├ sub-001_ses-01_task-control_run-01_bold.nii.gz / .json
│   │ ├ sub-001_ses-01_task-fmri_run-01_sbref.nii.gz / .json
│   │ ├ sub-001_ses-01_task-fmri_run-01_events.tsv
│   │ └ sub-001_ses-01_task-rest_run-01_recording-pulse_physio.tsv.gz / .json
│   ├ dwi/
│   │ ├ sub-001_ses-01_run-01_dwi.nii.gz / .json
│   │ └ sub-001_ses-01_run-01_dwi.bval / .bvec
│   ├ fmap/
│   │ ├ sub-001_ses-01_dir-AP_run-01_epi.nii.gz / .json
│   │ └ sub-001_ses-01_dir-PA_run-01_epi.nii.gz / .json
│   └ sub-001_ses-01_scans.tsv
├ derivatives/
│ ├ mriqc/
│ └ tractoflow_normalize_dwi/
├ code/
│ ├ task-movie/
│ └ task-fmri dictionaries / scripts
└ docs/
  ├ inventory / events / physiology
  └ deidentification / bids_validation / QC
```

Figure 1: Overview of the structure of the released multimodal MRI dataset. The dataset is organised according to the Brain Imaging Data Structure (BIDS), with anatomical, functional, diffusion, and field-map data organised within participant and session directories. Task events, physiological recordings, quality-control outputs, processing derivatives, analysis code, and supporting documentation are also provided.

(including presentStimParams.m, task-fmri_condition_dictionary.tsv, task-fmri_condition_dictionary.json, and code/task-movie/). Supporting documentation under docs/ covers inventory, events, physiology, de-identification, BIDS validation, and quality-control summaries. The file .bidsignore excludes *_TB1TFL* and *_localizer* series from BIDS validation while leaving those files present in the tree.

Each packaged session follows the layout sub-<id>/ses-<id>/, with modality folders anat/, func/, dwi/, and fmap/ when the corresponding data were collected, and a session-level sub-<id>_ses-<id>_scans.tsv listing imaging files in that session (acq_time is recorded as n/a). Imaging volumes are distributed as gzip-compressed NIfTI (*.nii.gz) with JSON sidecars. Anatomical data under anat/ include T1-weighted volumes labelled *_T1w.nii.gz. Sidecar ProtocolName/SeriesDescription distinguish standard MPRAGE series (T1w_MPR; 266 volumes in the release) from white-matter-nulled MPRAGE (WMn_MPRAGE_sagittal; 121 volumes). Three-dimensional FLAIR volumes (*_FLAIR.nii.gz; 137 files) are also released. Turbo-flash B1-mapping series (*_TB1TFL.nii.gz; 272 files) and localizer images are present under anat/ but are listed in .bidsignore. Released T1w and FLAIR anatomicals in the BIDS anat/ hierarchy are the defaced volumes used for sharing; a separate derivatives/defacing/ tree is not duplicated inside this package.

Functional data are stored under func/ as magnitude BOLD series (*_bold.nii.gz) for four task labels: task-rest, task-movie, task-fmri, and task-control. Coverage is incomplete across sessions: the release contains 135 magnitude task-rest runs (132 sessions; typically one run per session), 536 task-movie runs (134 sessions; most often four runs), 553 task-fmri runs (134 sessions; most often four runs), and 401 task-control runs (135 sessions; most often three runs). Representative volume counts in archived magnitude series are 320 (task-rest), 210 (task-movie), 226 (task-fmri), and 20 (task-control); individual runs may differ. Matching single-band reference images (*_sbref.nii.gz) and phase images (part-phase) are included for many functional acquisitions. Event tables (*_events.tsv) accompany all 536 magnitude task-movie runs and 514 of the 553 magnitude task-fmri runs; task-rest and task-control have no event files by design. Movie events include onset, duration, trial_type, stim_file, eye, movie_label, and matlab_run_id. Task-fMRI events provide block onsets and trial_type labels (baseline or stim-01–stim-12), with condition definitions in the root and code/ dictionaries. Copyrighted movie video files are not redistributed; clip identity is recorded in the event tables and code/task-movie/. Spin-echo EPI field maps under fmap/ use opposing phase-encode entities dir-AP and dir-PA (*_epi.nii.gz; 268 anterior–posterior and 267 posterior–anterior magnitude series in the release, present in all 135 sessions) and are intended for susceptibility distortion correction of EPI data. Field-map sidecars include phase-encoding direction and, where available, IntendedFor references to associated functional or diffusion series.

Diffusion MRI under dwi/ is released as *_dwi.nii.gz with matching JSON sidecars and FSL-style *.bval/*.bvec gradient tables (911 magnitude diffusion series across 131 sessions). Protocol families in the sidecars include gSlider SMS acquisitions (gsld_76dir_b2000_1mmiso_AP, multi-shell $b = 0/1000/2000$ s/mm^2; and short posterior–anterior $b = 0$ series such as gsld_75TE_PA_3b0) and RESOLVE trace acquisitions (resolve_3scan_trace_tra_p3_160_1.4is and resolve_3scan_trace_tra_p3_160_1.4iso_PA, including TRACEW series; $b = 0/1000$ s/mm^2 in representative tables). Phase-encoding direction is recorded in the JSON sidecars and varies across runs.

Physiological recordings, when conversion succeeded, are stored alongside the corresponding functional runs as BIDS physiology files *_recording-<channel>_physio.tsv.g with JSON sidecars. Channels present in the release are pulse (1127 files), respiratory (1258), and trigger (1223), totalling 3608 physiology time-series files spanning 77 participants and 116 sessions; absence of physiology for a given run does not by itself indicate invalid BOLD data. Derivatives included under derivatives/ are (i) mriqc/, containing native MRIQC image-quality metric JSON for all 84 participants (HTML reports are not packaged), and (ii) tractoflow_normalize_dwi/, containing TractoFlow Normalize_DWI outputs for 66 participants as sub-<id>/ses-01/dwi/sub-<id>_ses-01_space-dw Visual diffusion QC materials associated with that derivative are provided under docs/quality_control/dwi/.

3. MRI Data Acquisition

3.1. MRI Scanner

3.2. Anatomical MRI

3.3. Functional MRI

3.4. Field Maps

3.5. Physiological Recordings

4. Experimental Paradigms

Table 3: MRI acquisition protocol: laboratory SeriesDescription mapped to BIDS datatype and general filename pattern (single session, console order). Run index <N> is assigned at conversion and may differ from acquisition order. Full form: sub-<ID>_ses-<01|02>_<entities>_<suffix>.nii.gz.

#	Lab series name	BIDS datatype	General BIDS name
1	Localizer	anat	run-<N>_localizer
2	fMRI1_AP	func	task-fmri_run-<N>_bold
3	fMRI2_AP	func	task-fmri_run-<N>_bold
4	Control1_PA	func	task-control_run-<N>_bold
5	fMRI3_AP	func	task-fmri_run-<N>_bold
6	fMRI4_AP	func	task-fmri_run-<N>_bold
7	Movie1_AP	func	task-movie_run-<N>_bold
8	Movie2_AP	func	task-movie_run-<N>_bold
9	SpinEchoFieldMap_AP	fmap	dir-AP_run-<N>_epi
10	SpinEchoFieldMap_PA	fmap	dir-PA_run-<N>_epi
11	Movie3_AP	func	task-movie_run-<N>_bold
12	Movie4_AP	func	task-movie_run-<N>_bold
13	REST1_AP	func	task-rest_run-<N>_bold
*			
14	Localizer	anat	run-<N>_localizer
15	T1w_MPR	anat	run-<N>_T1w
16	Control2_AP	func	task-control_run-<N>_bold
17	Control3_PA	func	task-control_run-<N>_bold
18	SpinEchoFieldMap_AP	fmap	dir-AP_run-<N>_epi
19	SpinEchoFieldMap_PA	fmap	dir-PA_run-<N>_epi
20	gsld_76dir_b2000_1mmiso_AP	dwi	run-<N>_dwi
21	gsld_75TE_PA_3b0	dwi	run-<N>_dwi
22	tfl_b1map_1mmiso	anat	run-<N>_TB1TFL
23	resolve_3scan_trace_tra_p3_160_1.4iso_AP	dwi	run-<N>_dwi
24	resolve_3scan_trace_tra_p3_160_1.4iso_PA	dwi	run-<N>_dwi
25	Sag Flair 3D-0.8	anat	run-<N>_FLAIR
26	WMn_MPRAGE_sagittal	anat	run-<N>_T1w

* Anterior part of the head coil added, after which the localizer is repeated.

4.1. Grating Task

4.2. Movie Paradigm

4.3. Control Condition

4.4. Resting-State Paradigm

5. Data Processing

Placeholder for Figure 2 — Data processing workflow.
Add `figures/Figure_2.pdf` and uncomment `\includegraphics`.

Figure 2: [Caption for DICOM-to-BIDS processing workflow.]

5.1. DICOM Organization

5.2. De-identification

5.3. DICOM-to-NIfTI Conversion

5.4. BIDS Conversion

5.5. Metadata Generation

6. Quality Control

Placeholder for Figure 4 — Quality control / completeness.
Add `figures/Figure_4.pdf` and uncomment `\includegraphics`.

Figure 3: [Caption for quality control and/or protocol completeness.]

Table 4: Protocol completeness.

Acquisition	Expected	Available	Missing
[TODO: Sequence / task]	[TODO]	[TODO]	[TODO]
[TODO: Sequence / task]	[TODO]	[TODO]	[TODO]
[TODO: Sequence / task]	[TODO]	[TODO]	[TODO]

[TODO: Populate from QC / inventory reports; define Expected vs Available.]

6.1. Automated Quality Control

6.2. Visual Quality Control

7. Data Records

Placeholder for Figure 3 — Dataset organization or technical validation.
Add `figures/Figure_3.pdf` and uncomment `\includegraphics`.

Figure 4: [Caption for dataset organization / BIDS layout overview.]

7.1. Dataset Organization

7.2. File Formats

7.3. Dataset Contents

7.4. Data Dictionary

8. Technical Validation

8.1. Structural Validation

8.2. Metadata Validation

8.3. Image Validation

8.4. Dataset Completeness

9. Usage Notes

9.1. Recommended Uses

This dataset is prepared for secondary analysis and does not present new biological findings. Documented intended reuse includes methods development, multimodal MRI analysis, quality-control benchmarking, and reproducible neuroimaging workflows that start from BIDS inputs. The release includes structural, functional, diffusion, and field-map MRI organized according to BIDS (BIDSVersion 1.9.0), with a longitudinal design of up to two sessions (`ses-01`, `ses-02`) when both visits exist.

9.2. Important Considerations

Users should account for the following documented constraints when planning analyses.

- **Longitudinal coverage is incomplete.** Of 84 subjects in the BIDS tree, 51 have both sessions, 32 have `ses-01` only, and 1 has `ses-02` only. Missing sessions reflect incomplete participant follow-up, not omitted files. BIDS Validator warnings such as `MISSING_SESSION` and `INCONSISTENT_SUBJECTS` are expected.

- **Task events are not available for every functional run.** For `task-fmri` (laboratory visual grating / checkerboard paradigm), event timing files accompany 514 of 553 magnitude BOLD series (92.9%); runs without an unambiguous protocol-to-acquisition mapping intentionally lack `events.tsv`. For `task-movie`, design-level `events.tsv` files are provided for all magnitude movie BOLD runs, but onsets follow a documented run-level / continuous-movie timing policy and are **not** millisecond TTL-locked onsets. Movie stimulus video files are **not** redistributed (third-party copyright); protocol code and timing metadata are included under `code/task-movie/`. For `task-rest`, no `events.tsv` are released by design (continuous fixation without discrete trials); validator `EVENTS_TSV_MISSING` warnings are expected. Control runs lack recoverable trial timing and intentionally have no `events.tsv`. Incomplete task coverage includes `sub-055` (no rest) and `sub-063` (no grating), as documented in the release README.

- **Public package scope.** The public package is intended as BIDS data, with defaced anatomical derivatives for sharing where applicable, and **must not** redistribute identifiable source DICOM (for example under `raw_original/`). Defaced structural volumes are generated under a derivatives tree; original BIDS anatomicals are not overwritten by defacing.

- **Derivatives versus raw acquisitions.** Preprocessed diffusion volumes under `derivatives/tractoflow_normalize_dwi/` were generated by an upstream workflow, are present for only a partial subset of subjects, and should be treated as derivatives—not as replacements for raw BIDS `dwi/` runs. Upstream analysis products such as HCP structural preprocessing outputs, Benson's Atlas labels, tractography, or connectivity matrices should not be assumed to be present unless listed as verified contents of the release tree.

- **Quality control.** Automated QC outputs (including BIDS validation, MRIQC where available, Pizarro T1w screening, DWI QC, defacing audit, and privacy audit) are used to describe and prioritize inspection. They are **not** automatic exclusion criteria for this release description. MRIQC and cohort-wide DWI QC summaries were not finished at the time of the current release README. Residual visual review of face removal remains recommended even where T1w defacing coverage is complete (356/356 paired T1w in the documented audit).

- **De-identification.** De-identification follows a DICOM PS3.15-oriented custom workflow and is **not** a claim of certified full compliance with the DICOM PS3.15 Basic Profile. Siemens PhysioLog objects that passed documented conversion gates are distributed as BIDS physiology sidecars; peripheral PMU logs and failed-gate PhysioLogs remain source-only.

- **License / access.** This dataset is intended for contractual release via FITBIR. Access is governed by FITBIR data-access agreements and study-specific sharing terms (see the dataset `LICENSE`). It is **not** released under CC0. Repository accession / how-to-cite strings remain to be inserted when the FITBIR deposit is finalized.

9.3. Software and Tools

Release documentation records the following software used to convert, organize, validate, or quality-check the dataset (versions as stated in the release README and related reports):

- `neuro_pipeline` (dataset_description GeneratedBy version 2.1.0) for DICOM-to-BIDS organization;
- dcm2niix for NIfTI + JSON conversion (sidecar provenance typically records `v1.0.20230411`);
- bids-validator 1.15.0;
- MRIQC (approximately version 24.x; coverage partial at the time of documentation);
- pydeface 2.0.0 (Compute Canada build) for anatomical defacing derivatives;
- Pizarro T1w screening with ONNX Runtime 1.23.2;
- MRtrix3 tools for DWI QC;
- Python 3.11.x in documented report environments.

Release-facing paradigm code and dictionaries live under `release_dataset/code/` (including `code/task-grating/`, `code/task-movie/`, and related condition dictionaries). Pipeline, cleaning, QC, and audit scripts are documented in the project workspace (examples listed in the release README under Code availability). Users working from BIDS inputs may also use standard BIDS-compatible neuroimaging tools of their choice; the items above are those recorded for the construction and validation of this release, not a mandatory analysis stack.

10. Limitations

11. Ethics Statement

12. Data Availability

Repository name:
Dataset DOI:
Version:
Access conditions:
License:
Restrictions:
Related software repository:

13. Author Contributions

Conceptualization:
Data curation:
Formal analysis:
Funding acquisition:
Investigation:
Methodology:
Project administration:
Resources:
Software:
Supervision:
Validation:
Visualization:
Writing – original draft:
Writing – review and editing:

14. Acknowledgements

15. Declaration of Competing Interest